

HASH Innovation Challenge Prototype Submission Requirements

To facilitate fair evaluation, reproducibility, and future scalability, all teams are required to submit a prototype package consisting of the following materials.

1. Prototype Demonstration

Teams must submit a functional prototype or proof-of-concept demonstrating the core features of their solution.

Acceptable formats include:

- Web application
- Mobile application
- Conversational AI chatbot
- Decision-support system
- Machine learning model
- Data analytics dashboard
- Other AI-enabled digital health tools

The prototype does not need to be production-ready but must demonstrate technical feasibility and intended functionality.

2. Technical Report (Maximum 10 Pages)

A technical report in PDF format describing:

A. Problem Statement

- Challenge track selected
- Reproductive health problem being addressed
- Target users and beneficiaries
- Relevance to Sub-Saharan African contexts

B. Solution Overview

- Description of the proposed innovation
- Key functionalities
- User workflow
- Expected health system benefits

C. Technical Approach

- AI/ML methods used
- Algorithms and models implemented

- System architecture
- Data processing pipeline
- Model evaluation methods

D. Ethical and Responsible AI Considerations

- Data privacy and security measures
- Bias and fairness considerations
- Transparency and explainability approaches
- Alignment with local/regional legal and regulatory requirements

E. Scalability and Sustainability

- Potential deployment strategy
- Resource requirements
- Integration with existing health systems
- Long-term maintenance considerations

3. Source Code Repository

Teams must provide access to a GitHub, GitLab, or equivalent source code repository containing:

- Complete source code
- Installation instructions
- Dependency specifications
- Environment configuration files
- Documentation for reproducing results

Repositories should remain accessible to organizers and judges throughout the evaluation period.

Required files include:

- README.md
- requirements.txt or environment.yml
- LICENSE file
- CONTRIBUTING.md (optional)

4. Data use Documentation

Teams must submit a data inventory describing:

A. Data Sources

- Challenge datasets used from the DASSA platform
- Public datasets integrated and any externally sourced datasets

B. Data Management

- Data cleaning procedures
- Feature engineering methods
- Data quality checks

C. Data Governance

- Data permissions and licenses for public and external datasets used.
- Privacy protection measures
- Compliance with ethical standards

5. Reproducibility Package

Teams must provide sufficient materials to allow reviewers to reproduce the main results.

This should include:

- Jupyter notebooks or scripts
- Model training workflow
- Evaluation scripts
- Sample input and output files
- Instructions for running the prototype

6. Team Contribution Statement

Teams must provide a brief summary of contributions from each member, including:

- Technical development
- Data science and analytics
- Clinical or public health expertise
- Project management
- User experience design

7. Intellectual Property and Open Science Declaration

Teams must submit a declaration indicating:

- Ownership of submitted work
- Use of open-source components where applicable
- Compliance with dataset licenses
- Permission for organizers to showcase submitted prototypes for educational and dissemination purposes

Submission Checklist

Before submission, ensure the following are included:

- Functional prototype or proof-of-concept
- Technical report (PDF)
- Source code documentation i.e. GitHub/GitLab repository link
- Data use documentation
- Reproducibility package
- Team contribution statement
- Intellectual property and open science declaration